25901889 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)

THIRD SEMESTER

Discipline Specific Core Course (DSC) - MG3DSECHE201 - INTRODUCTION TO NANOSCIENCE

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **5** questions

Each question carries **4** marks

1. How do magnetotactic bacteria utilize magnetic nanoparticles to orient themselves in their environment? [U] / [CO1]
2. What are the different type of signals produced when an electron beam interacts with sample in electron microscopy? [U] / [CO2]
3. Define magnetic nanoparticles and give two examples. [K] / [CO3]
4. What are the benefits of using nanoparticles in medicinal therapy? [K] / [CO4]
5. Describe different types of nanomaterials based on their dimensionality. [K] / [CO1]
6. Explain the role of carbon dots in bioimaging. [U] / [CO3]
7. How do nanomaterials facilitate data storage? [An] / [CO4]

[4x5 = 20]

Part B

Short Essay Type Questions

Answer any **5** questions

Each question carries **7** marks

8. Describe the key characteristics of metal oxide nanoparticles and their significance. [U] / [CO3]

9. How is nanotechnology applied in hydrogen storage materials and devices? [A] / [CO4]

10. How does nanotechnology help to reduce pollutants in air pollution control systems? [U] / [CO4]

11. Explain the different types of nanomaterials:
(a) carbon-based, (b) metal-based, (c) dendrimers, and (d) composites.
Comment on their main properties and applications. [U] / [CO1]

12. Compare the advantages of SEM and TEM. [U] / [CO2]

13. Write the working principle and procedure of chemical precipitation method for nanomaterial synthesis. How pH, temperature and precursor concentration influence particle size and morphology of synthesised nanomaterials? [U] / [CO1]

14. What is X-ray Photoelectron Spectroscopy. Explain its importance in nanotechnology. [U] / [CO2]

[7x5 = 35]

Part C

Essay Type Questions

Answer any **1** questions

Each question carries **15** marks

15. Explain Transmission Electron Microscopy discussing its principle, applications, advantages and basic instrumental features. [U] / [CO2]

16. Explain the top down approaches used in nanochemistry. Discuss its methods, principle, advantages and limitations with suitable examples. [U] / [CO1]

[15x1 = 15]